

II. Let x candidates be interviewed by C.

Number of candidates interviewed by A = $(x + 2)$.

Number of candidates interviewed by B = $(x + 1)$.

$$\therefore x + (x + 2) + (x + 1) = 45 \Leftrightarrow 3x = 42 \Leftrightarrow x = 14.$$

So, the number of candidates interviewed by panel A is 14.

Thus, II alone is sufficient. Hence, the correct answer is (b).

- 21.** Let $P = x$, $Q = x + 2$, $R = x + 4$ and $S = x + 6$.

I gives, $\frac{x + (x + 2) + (x + 4) + (x + 6)}{4} = 11$

$$\Rightarrow 4x + 12 = 44$$

$$\Rightarrow 4x = 32 \Rightarrow x = 8.$$

$$\therefore \text{Largest integer} = x + 6 = 14.$$

So, I alone gives the answer.

II gives, $\frac{x + 6}{x} < 10 \Rightarrow x + 6 < 10x \Rightarrow 9x > 6 \Rightarrow x > \frac{2}{3}.$

Thus, II alone does not give the answer.

$\therefore$ Correct answer is (a).

- 22.** The average score depends on all the individual scores and the range of scores as given in I and II, is insufficient to calculate it.

So, both I and II together are insufficient.

$\therefore$ Correct answer is (d).

- 23.** **I** gives, sum of all 8 scores = $(84 \times 8) = 672$ (i)

II gives, sum of remaining 6 scores = $(86 \times 6) = 516$ (ii)

From (i) and (ii), we have:

$$\text{Sum of best and worst scores} = 672 - 516 = 156.$$

$$\therefore \text{Average of best and worst scores} = \frac{156}{2} = 78\%.$$

Thus, both I and II together give the answer. $\therefore$ Correct answer is (e).

- 24.** **II** gives, Average of marks in Mathematics, Science, English and Hindi = $\frac{250}{4} = 62.5$.

$$\therefore \text{Marks in Geography} = (62.5 + 16) = 78.5.$$

So, II alone gives the answer while I alone is insufficient.

$\therefore$ Correct answer is (b).

- 25.** To calculate the average attendance per game, we need to know the total attendance and the number of games played.

I and **II** give:

Combined average attendance per game

$$= \left(\frac{23100 + 25200}{20 + 22} \right) = \frac{48300}{42} = 1150.$$

So, both I and II together are needed.

$\therefore$ Correct answer is (e).

- 26.** Average age of 11 students and 1 teacher = 14 years

$$\Rightarrow \text{Total age of (11 students and 1 teacher)} = (14 \times 12) \text{ years} = 168 \text{ years}.$$

$$\text{Average age of (11 students and 1 teacher)} = (\text{Average age of 11 students}) + 3$$

$$\Rightarrow \text{Average age of 11 students} = (14 - 3) \text{ years} = 11 \text{ years}$$

$$\Rightarrow \text{Total age of 11 students} = (11 \times 11) \text{ years} = 121 \text{ years}.$$

$$\therefore \text{Age of the teacher} = (168 - 121) \text{ years} = 47 \text{ years}.$$

Thus, both I and II are needed to get the answer.

$\therefore$ Correct answer is (e).

- 27.** **I.** Total weight of 46 children = $[(22.5 \times 30) + (29.125 \times 16)] \text{ kg} = 1141 \text{ kg}.$

Weight excluded is not exact. So, average of remaining class cannot be obtained.

II. Total weight of 45 children = $[(23.5 \times 46) - 46] \text{ kg} = 1035 \text{ kg}.$

$$\text{Average weight of 45 children} = \frac{1035}{45} \text{ kg} = 23 \text{ kg}.$$

$\therefore$ Data in II is sufficient to answer the question, while the data in I is not sufficient.

$\therefore$ Correct answer is (b).

- 28.** Suppose the secretary's speech lasted for x minutes.

I gives, number of words in the speech = $50x$.

II gives, number of words in the speech = $46(x + 10)$.

$$\therefore 50x = 46(x + 10) \Rightarrow 4x = 460 \Rightarrow x = 115.$$

So, the secretary's speech lasted for 115 min *i.e.*, 1 hr 55 min.

Thus, both I and II are needed to find the answer.

$\therefore$ Correct answer is (e).

- 29.** **I** gives, total marks in 4 subjects = $(60 \times 4) = 240$ (i)

II gives, $E + M = 170$ (ii)

III gives, $M + S = 180$.

Thus, none of (a), (b), (c), (d) is true.

$\therefore$ Correct answer is (e).

- 30.** **I** gives, sum of weights of 60 students = $(42 \times 60) \text{ kg} = 2520 \text{ kg}$ (i)

II gives, if there be x boys in the class, sum of weights of x boys = $(43x) \text{ kg}$ (ii)

III gives, sum of weights of all the girls = 1144 kg (iii)

From (i), (ii), and (iii), we have:

$$43x + 1144 = 2520 \Rightarrow 43x = 1376 \Rightarrow x = 32.$$

$$\text{So, number of boys} = 32, \text{ number of girls} = (60 - 32) = 28.$$

$$\therefore \text{Average weight of girls} = \left(\frac{1144}{28} \right) \text{ kg} = 40.86 \text{ kg}.$$

Thus, all I, II and III are needed to find the answer.

$\therefore$ Correct answer is (c).

- 31.** From **I** and **II**, we have:

Let the number of boys and girls be $4x$ and $5x$ respectively.

$$\text{Then, } 4x + 5x = 45 \Rightarrow 9x = 45 \text{ or } x = 5.$$

$$\text{So, number of boys} = 20; \text{ number of girls} = 25.$$

$$\therefore \text{Average age of 45 students} = \left(\frac{15.4 \times 20 + 14.6 \times 25}{45} \right) \text{ yrs}$$

$$= \left(\frac{673}{45} \right) \text{ years} = 14.96 \text{ years}.$$

Thus, I and II together give the answer.

From I and III, we have

$$\text{Let number of girls be } x. \text{ Then, number of boys} = x - 5.$$